

A pre-Feasibility Study for Establishing of Leather Clothes Plant.



2017



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1. Executive summary

The idea of the project is to manufacture leather clothes that are in line with the finest designs and focusing on men's and women's coats.

The idea was presented because of the availability of skilled-labour in sewing and manufacturing clothes in the governorate, and the high profitability of such projects. It is worth mentioning that to ensure the success of the project, it is necessary to know the requirements and consumer needs in the local and foreign markets.

The following table shows the economic indicators of the project:

Table1 Economic indicators

Indicator	Value
Over all cost of the Investment (JD)	1,182.178
Internal Rate of Return (IRR)	%21.58
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	798,194
Job Opportunities created	39
Total Revenues (For 10 Years)	9,629.659
Total Operating Expenses (For 10 years)	6,317.601
Total Net Profit (For 10 years)	1,319.513

Key Highlight of Jordan

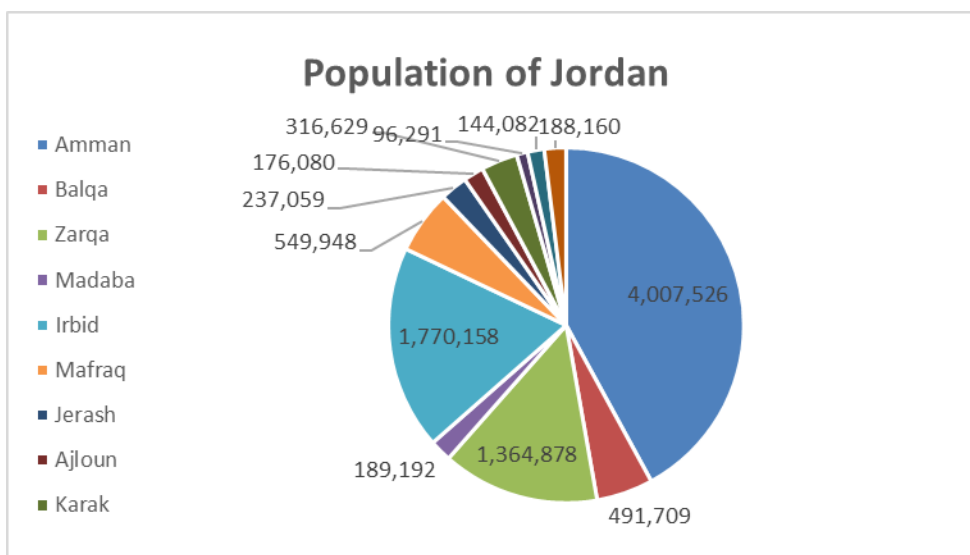
Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.



Of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

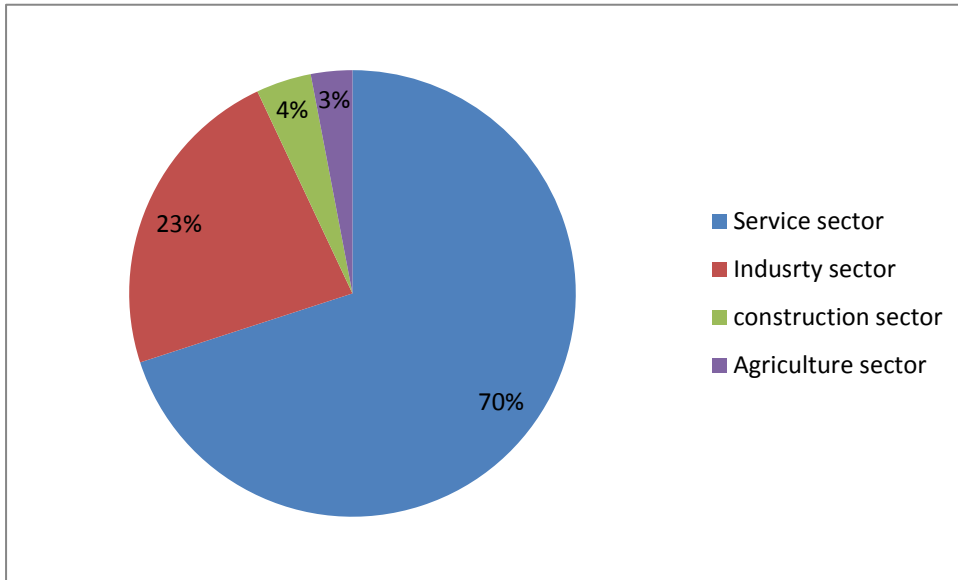
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Jarash:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table3 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20% As per Investment Law categorization, Irbid is classified in Class (C) where investors enjoys 60% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

2. Market Analysis

Project Description

The idea of the project is to manufacture leather clothes that are in line with the finest designs and focus on men's and women's coats.

The idea came about because of the availability of skilled labour in sewing and manufacturing of clothes, in the governorate the high profitability of such projects. To ensure the success of the project, it is necessary to know the requirements and consumer needs in the local and foreign markets.

Project Objectives

The proposed project aims to:

- Achieve a profitable return for project owners.
- Produce of leather garments with attractive designs that meet the needs of the Jordanian market and the external markets.
- Create job opportunities and reduce unemployment.

Proposed Product

The production of leather garments in line with the finest designs with special focus on men's and women's coats.

Target Markets

- Jordanian local market.
- External market.

Market size

The leather and garment sector is one of the most influential sectors in the Jordanian economy. Local exports of this sector reached 20.8% of total industrial exports in 2015. Leather tanning is a key sub-sector. Total raw leather exports of the Kingdom amounted to \$ 6.5 million according to the International Trade Center (ITC). Raw skins can be manufactured in Jordan and exported in the form of finished products, which increases the value added of this industry.

The increase in global and local demand for leather goods is one of the reasons that gives leather industries a great importance. Leather goods are considered a daily need Jordan's population is growing continuously, reaching 9,59 million people in 2015 (Appendix 1, population).

The following table shows the Kingdom's imports of all types of leather:

Table 4 The Kingdom imports of leather (inUSD)

Type	2011	2012	2013	2014	2015
Raw leather (except leather with furs) and tanning	510,000	584,000	596,000	674,000	694,000
Leather goods handbags,,etc.	18,653,000	25,689,000	26,542,000	30,702,000	31,741,000
Leather with natural fur.	45,000	42,000	61,000	396,000	195,000
Footwear and similar products	54,909,000	59,878,000	73,102,000	65,821,000	69,448,000

The following table shows the Kingdom's leather export:

Table 5 the Kingdom's exports of leather (in USD)

Type	2011	2012	2013	2014	2015
Raw leather (except leather with furs) and tanning	6,707,000	4,692,000	6,442,000	7,344,000	6,699,000
Leather goods, handling,,etc.	12,137,000	10,978,000	3,663,000	2,105,000	1,811,000
Leather with natural fur.	319,000	472,000	544,000	653,000	852,000
Footwear and similar products	14,247,000	27,127,000	7,886,000	3,731,000	4,289,000

Competators Analysis

Despite the existence of leather goods companies, there is no company specialized in leather clothing.

Leather goods companies in Jordan are:

- **Al-Eman Company:** Leather and wool trade, located at Wasfi Al-Tal Street in Amman.
- **Hijazi & Ghousheh Company:** Specializes in leather, Livestock, general trade, food stuff, trading agencies, establishment of farms.
- **Arab Co. for Leather & Wool Trading:** Specializes in food, agricultural, and animal industries (except fish). It also specializes in animal wool and skins. It is located in Zarq.

Price Analysis and Pricing policy

The Price of high quality natural leather JD10-12, per 1 meter. To make a leather coat, 2-3 meters of leather is required. Accordingly, the cost of producing natural leather coat ranges from JD 20-36.

The following table shows the price assumptions:

Table 6 prices assumptions

Price assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price of item	120	124	127	131	135	139	143	148	152	157

Market share

It was assumed that 7000 would be produced per year. Since each coat requires 2-3 meters of natural leather, the required amount for annual production is 17,500 meters. The amount of leather required for coats is 20% of the Kingdom's total imports of leather used in different industries. This is equivalent to 163,900 items, representing 4.3% of the Kingdom's leather imports.

Projected revenues

Projected annual revenue for the project

Table 7 projected revenues during (2018-2027)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Number of items	7000	7000	7000	7000	7000	7000	7000	7000	7000	7000
Revenues	840000	865200	891156	917891	945427	973790	1003004	1033094	1064087	1096009

3. Technical Analysis

This section explains the technical specifications of the project, and provides an assessment and description of the investment and operating costs based on the results of the market study.

The technical analysis also includes a general description of the project and its components, in addition to the construction and operational requirements and an identification of the aspects based on which the technical team assessed the costs.

The required human resources

The following table shows the human resources needed for the project:

Table 8 the required labor for the project

Job title	No.
General manager	1
Accountant	2
production line supervisor	1
Warehouse supervisor	1
Export supervisor	1
Guard	1
Total of indirect staff	8
Purchases officer	2
Packaging officer	2
Ironing officer	4
Clothing designers	3
Quality auditor	2
Maintenance technicians	15
cleaners	3
Total of direct staff	31
Total	39

Job descriptions of manpower:

- **General Manager:** Works independently within the scope of authority delegated to him. The GM proposes policies and work plans, , manages work in the factory, follows up on the optimal implementation of plans and programs in accordance with the policies, goals and objectives , oversees the administrative and financial management, manages subordinates and the development of their skills, as well as the development of procedures and ensures that occupational and public safety and health requirements are met.
- **Accountant:** Follows up on financial matters and financial audit, and submits annual reports on the financial situation.
- **Production line supervisor:** Oversees the production lines according to the applied policies and procedures and ensures the smooth operation of production during shifts.
- **Warehouse supervisor:** Implements plans and organizes all matters related to warehouses (raw materials, final product, product under production and storage areas), reports to the warehouse manager and organizes the entry and exit of the stock according to the procedures.
- **Export supervisor:** Monitors import and export processes and procedures of in accordance with the policies and plans set for this purpose, evaluates progress, monitors the fluctuation of commodity prices, provides the required goods with the required quality and the lowest prices to ensure their competitiveness in the local and foreign markets.
- **Security guard:** Implements the approved security instructions to maintain the security system in the factory.
- **Quality auditors:** Check the effectiveness of administrative performance and work procedures in accordance with the applicable legislation .
- **Maintenance technicians:** implement actions and activities of the electrical and mechanical maintenance, and follow up on the implementation of plans and programs in order to ensure the continuity of the production process and best quality at at the lowest possible cost, and ensure the availability of spare parts, and the preparation of technical reports.
- **Purchases officer:** Organize, and maintain records and forms of procurement management and assisting the purchasing manager in his duties, as required.
- **Ironing officer:** irons items.
- **Clothing designers:** Designs the required items and follow-up on their execution.

- **Cleaner:** Cleaning, polishing and disinfection works for senior management offices, cleaning of office rooms, office surfaces, bathrooms, entrances and stairs.

Site analysis

The proposed location for establishing the factory is Al-Mustaba district in Jerash, because it is close to Zarqa governorate, which has a large number of untreated, natural leather warehouses, and the labor necessary to operate the plant.

Technical requirements analysis

Land and construction

The project will be constructed on an area of 3,000 sq.m, and costs JD 21,000 in Al-Mustaba in Jerash governorate. The land will be divided into 2,000 sq.m for plant construction, 500 sq.m for the warehouse, 400 sq.m for vehicles and means of transportation and 100 sq.m for administrative offices.

The following table shows the plot of land required for the project:

Table9 The required land for the project

Land	Area sq.m
The required land	3,000

The required construction works for the project are as follows:

Table 10 The required construction work

Construction Work	Area (sq.m)
Constructions works	2600
Design and blueprints	500
Excavation work	3000
Paving	400
Soil examination	-

Machinery and equipments

The following table shows the machinery and equipment needed for the project:

Table 11 Machinery and equipments

Machinery and equipment	Quantity
Fire extinguishers	1
Central air conditioner	1
Stitching machine	30
Leather merging machines	1
Cautery	2
Leather cutting machines	2
Tables for checking items	3
Machines for installing buttons	1
Machines for installing fasteners	1
Punching machines	1
Storage racks	10

Legal Requirements

Ministry of Industry and Trade

The establishment of a company and opening a registration and a commercial name are done through the ministry of industry and trade. Therefore, it is necessary to bring the registration certificate of the company and the lease contract and identification card in order to apply for a professions license and fill out the following forms:

- Registration with the Industry and Trade Chamber
- Bank deposit letter.
- Occupational permit which requires the blueprints of the site on which the project will be constructed and the approval of Jerash Municipality.

Approval must be obtained from the municipality and the following fees must be paid:

- Profession fees: X percentage of capital (determined by the municipality).
- JD15 advertising fee.
- Waste fees 20% of the profession fees.
- Health fees.

Ministry of Environment: To apply for the approval of the Ministry of Environment, the following documents should be submitted:

مشروع مصنع ملابس جلدية- محافظة جرش

- The coordinates of the land.
- Registration certificate (from industry and trade).
- Land blueprint.
- Organization chart.

The documents are sent to the Environment Directorate in Jerash governorate, then transferred to the Ministry of Environment. After that, a request is submitted to the Ministry of Environment and the technical form is filled. The documents are then presented to the Licensing Committee.

Chamber of Industry: Registration in the Jarash Chamber of Industry shall be done by bringing the Commercial Register, the lease contract and the identity of the person concerned. The fees are determined according to the capital of the project. A JD 600 in case the capital is between JD (500,000 and 999,999).

The required fees for services:

- The amount of JD (100) is required when requesting for establishing new industrial projects or when adding a new production line.
- The amount of JD (100) at the time of inspection of the factory or plant or a new production line and control of production processes when issuing a disclosure decision and attaching the receipt with the decision.

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.61% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Table 12 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	35%
Utilities Expenses from Total Revenues	2.5%
Maintenance Expenses from Total Revenues	1%
Depreciation Expenses from Total Revenues	1%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.5%

Table 16 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	%5
(2) Income Tax Deduction	%0

Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	%10
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	5%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 17 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	5%
Bets	0.89
Equity	472,871
Loans Value	709,307
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	9.01%
Loans Value	709,307
Loans Percentage	60%
Equity	
Cost of Equity	%15.5
Equity Value	472,871
Equety Percentage	%40

Gross	
Project Value	1,182,178
Weighted Average Cost of Capital	%11.61

Capital Expenditures

The estimated cost of the project 1,182,178JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 18 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (m2)	Cost
Required Land	7	3,000	21,000
Cost			21,000

Table 19 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	275		715,000
Design and blueprints	7		3,500
Excavation	5		15,000
Paving	12		4,800
Soil Test	-	-	200
Other Works			36,925
Total			
Construction Works			775,425

Table 20 machinery and equipments Capital Expenditure

Description	No.	Cost (JOD per squared)	Gross Cost
Fire fighting equipment	1	500	500
Central air conditioner	1	35,000	35,000
Stitching machine	30	400	12,000
Leather merging machines	1	800	800
Cautery	2	40	80
Leather cutting machines	2	400	800

Tables for checking items	3	80	240
Machines for installing buttons	1	400	400
Machines for installing fasteners	1	400	400
Punching machines	1	1,000	1,000
Storage racks	10	15	150
Fire fighting equipment			51,370

Table 21 Cost of furniture required for the project

Description	No.	Cost (JOD per unit)	Gross cost
Furniture	-	-	12,000
Total			12,000

Table 22 Transportation and Vehicles Capex

Description	No.	Cost (JOD per unit)	Gross cost
Pick up	2	23,400	46,800
Total			46,800

Table 23 Pre Operating Expenses

Pre-operating expenses	الكلفة التقديرية
Governmental fees	3,000
Transportation	750
Marketing	2,500
Training	2,500
Total	8,750

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	135,465
General and administrative expenses	3	%25	3,047
Marketing expenses	3	%25	1,050
Indirect Salaries	3	%25	19,800
Total			159,362

Table 25 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	21,000	10%	23,100	%2.0
Construction works	775,425	10%	852,968	%72.2
Machinery & Equipment	51,370	10%	56,507	%4.8
Furniture	12,000	10%	13,200	%1.1
Vehicles	46,800	10%	51,480	%4.4
Pre-Operating Expenses	8,750	10%	9,625	%0.8
Working capital	159,362	10%	175,298	%14.8
Total (JD)	1,074,707		1,182,178	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,182,178) JD, where 60% of the project will be financed through loans (709,307) JD, and the remaining (40%) through (Equity) with (472,871) JD. The following table summarizes the general structure of the required financing:

Table 26 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	472,871	%40
Loans	709,307	%60
Total	1,182,178	%100
Use of Fund:	Value	Ratio
Capital Expenditures	997,255	%84.36
Pre-Operating Expenses	9,625	%0.81
Working Capital	175,298	%14.83
Total	1,182,178	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 27 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	294,000	302,820	311,905	321,262	330,900	340,827	351,051	361,583	372,430	383,603
Utilities from Revenues	21,000	21,630	22,279	22,947	23,636	24,345	25,075	25,827	26,602	27,400
Maintenance from Revenues	8,400	8,652	8,912	9,179	9,454	9,738	10,030	10,331	10,641	10,960
Disposables from Revenues	8,400	8,652	8,912	9,179	9,454	9,738	10,030	10,331	10,641	10,960
Direct Salaries	160,800	167,393	174,256	181,400	188,838	196,580	204,640	213,030	221,764	230,857
Others	49,260	50,915	52,626	54,397	56,228	58,123	60,083	62,110	64,208	66,378
Total	541,860	560,061	578,889	598,364	618,510	639,350	660,909	683,213	706,287	730,159

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 28 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	250	258	265	273	281	290	299	307	317	326
Stationary and Printing	250	258	265	273	281	290	299	307	317	326
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Training	700	721	743	765	788	811	836	861	887	913
Insurance	7,479	7,704	7,935	8,173	8,418	8,671	8,931	9,199	9,475	9,759
Miscellaneous	1,108	1,141	1,175	1,211	1,247	1,284	1,323	1,363	1,404	1,446
Total	12,187	12,553	12,930	13,317	13,717	14,128	14,552	14,989	15,439	15,902

Marketing Expenses

Table 29 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	4,200	4,326	4,456	4,589	4,727	4,869	5,015	5,165	5,320	5,480

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Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 30 Expected Annual Salaries and Wges (2027-2018)

Job title	No.	Monthly salary (JD)	Total monthly	Total annual wages
General manager	1	1,750	1,750	28,000
Accountat	2	450	900	14,400
supervisor for production line	1	800	800	12,800
Warehouse administrator	1	400	400	6,400
Export administrator	1	500	500	8,000
security	1	200	200	3,200
Indirect employment	8		4,950	79,200
purchase officer	2	400	800	12,800
cleaner	2	200	400	6,400
Cui workers	4	200	800	12,800
Clothing designers.	3	600	1,800	28,800
Quality audit staff	2	500	1,000	16,000
Machinery workers	15	300	4,500	72,000
Packaging worker	3	250	750	12,000
Dicrect employment	31		10,050	160,800
Total	39		10,050	240,000

The following table shows the expected annual salaries to employees for 10 years of the life of the project:

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employment										
General manager	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Accountat	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
supervisor for production line	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Warehouse administrator	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Export administrator	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
security	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
General manager	3,200	3,331	3,468	3,610	3,758	3,912	4,072	4,239	4,413	4,594
Indirect employment	79,200	82,447	85,828	89,346	93,010	96,823	100,793	104,925	109,227	113,706
Direct employment										
purchase officer	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
cleaner	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Cui workers	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Clothing designers.	28,800	29,981	31,210	32,490	33,822	35,208	36,652	38,155	39,719	41,347
Quality audit staff	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Machinery workers	72,000	74,952	78,025	81,224	84,554	88,021	91,630	95,387	99,298	103,369
Packaging worker	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Dicrect employment	160,800	167,393	174,256	181,400	188,838	196,580	204,640	213,030	221,764	230,857
Total	240,000	249,840	260,083	270,747	281,847	293,403	305,433	317,956	330,992	344,562
Depreciations										

The following tables illustrate cost of Capex and Depreciation rates:

Table 31 Capex and Depreciation Expenses

Equipment costs and capital expenditure, annual consumption rates	Cost (JOD)	Consumption ratio	Annual additions
Land	23,100	%0.0	%0.0
Construction Works	852,968	%5.0	%0.0
Machinaries	56,507	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0
Gross Total (JOD)	997,255		

The table below shows a summary of capital costs, additions and annual consumption

Table 32 Depreciations and Additions on Construction Works

Summary of capital costs and additions and consumption annually	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	997,2	999,0	1,000,8	1,002,7	1,004,7	1,006,7	1,008,8	1,011,0	1,013,2	1,015,5
Total Depreciation Expenses	54,76	54,94	55,131	55,321	55,517	55,720	55,929	56,145	56,367	56,597
Total Accumulated Depreciation	54,76	109,7	164,844	220,165	275,682	331,402	387,331	443,476	499,843	556,440
Total Additions	0	1,790	1,846	1,903	1,963	2,026	2,090	2,157	2,227	2,299
Total Net Book Values	942,4	889,3	836,047	782,629	729,075	675,381	621,542	567,554	513,414	459,116

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 33 Loan Details

Loan value	709,307
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2017
Annual Payment	184,634
Number of Payments	5

The following table shows the loan payments:

year	Annual Payment	Interest	Capital	The remainder of the loan
2017				709,307
2018	184,634	67,242	117,392	591,915
2019	184,634	56,114	128,521	463,394
2020	184,634	43,930	140,704	322,690
2021	184,634	30,591	154,043	168,646
2022	184,634	15,988	168,646	0

Income Statement

Through all the information and financial analysis of the project's financial expenses have statment income for the years (2018 – 2027) as shown in the following table:

Table 34 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	840,000	865,200	891,156	917,891	945,427	973,790	1,003,00	1,033,09	1,064,08	1,096,00
Gross Operating Revenues	840,000	865,200	891,156	917,891	945,427	973,790	1,003,00	1,033,09	1,064,08	1,096,00
Operating Expenses	541,860	560,061	578,889	598,364	618,510	639,350	660,909	683,213	706,287	730,159
Gross Operating Profit	298,140	305,139	312,267	319,527	326,918	334,440	342,095	349,881	357,800	365,851
<i>Gross Profit Percentage</i>	%35	%35	%35	%35	%35	%34	%34	%34	%34	%33
Salaries and Benefits (Indirect Staff)	(79,200)	(82,447)	(85,828)	(89,346)	(93,010)	(96,823)	100,793	104,925	109,227	113,706
General and Administraive Expenses	(12,187)	(12,553)	(12,930)	(13,317)	(13,717)	(14,128)	(14,552)	(14,989)	(15,439)	(15,902)
Markeing Expenses	(4,200)	(4,326)	(4,456)	(4,589)	(4,727)	(4,869)	(5,015)	(5,165)	(5,320)	(5,480)
Pre Operating Expenses	(9,625)									
Gross Indirect Expenses	105,212	(99,326)	103,213	107,253	111,454	115,820	120,360	125,080	129,986	135,087
Income before Interest, Depreciation and	192,928	205,812	209,054	212,273	215,464	218,620	221,735	224,802	227,814	230,764
Fixed Assets Depreciations	(54,767)	(54,946)	(55,131)	(55,321)	(55,517)	(55,720)	(55,929)	(56,145)	(56,367)	(56,597)
Income before Tax and Interests	138,161	150,866	153,924	156,952	159,946	162,900	165,806	168,657	171,447	174,166
Bank Interests	(67,242)	(56,114)	(43,930)	(30,591)	(15,988)	(0)	(0)	(0)	(0)	(0)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	70,918	94,753	109,994	126,361	143,959	162,900	165,806	168,657	171,447	174,166
Income Tax	(3,546)	(4,738)	(5,500)	(6,318)	(7,198)	(8,145)	(8,290)	(8,433)	(8,572)	(8,708)
Net Profit	67,372	90,015	104,494	120,043	136,761	154,755	157,515	160,224	162,874	165,458
<i>Net Profit Percentage</i>	<i>%8</i>	<i>%10</i>	<i>%12</i>	<i>%13</i>	<i>%14</i>	<i>%16</i>	<i>%16</i>	<i>%16</i>	<i>%15</i>	<i>%15</i>
Compulsory Reserves	(6,737)	(9,002)	(10,449)	(12,004)	(13,676)	(15,475)	(15,752)	(16,022)	(16,287)	(16,546)
Retained Earnings	60,635	141,649	235,694	343,733	466,817	606,097	747,861	892,062	1,038,64	1,187,56

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 35 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	67,372	90,015	104,494	120,043	136,761	154,755	157,515	160,224	162,874	165,458
Bank Interests	67,242	56,114	43,930	30,591	15,988	0	0	0	0	0
Depreciation	54,767	54,946	55,131	55,321	55,517	55,720	55,929	56,145	56,367	56,597
Total Operating Cashflows before Additions to Working Capital	189,382	201,075	203,555	205,955	208,266	210,475	213,444	216,369	219,242	222,055
Inventory (Increase/Decrease)	(24,500)	(735)	(757)	(780)	(803)	(827)	(852)	(878)	(904)	(931)
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	41,310	1,564	1,624	1,686	1,751	1,819	1,889	1,962	2,038	2,117
Working Capital (Increase/Decrease)	16,810	829	867	907	948	992	1,037	1,084	1,134	1,185
Net Cashflows from Operating Activities	206,192	201,903	204,421	206,862	209,214	211,466	214,481	217,453	220,375	223,241
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(997,255)	(1,790)	(1,846)	(1,903)	(1,963)	(2,026)	(2,090)	(2,157)	(2,227)	(2,299)
Net Cashflows from Investments Activities	(997,255)	(1,790)	(1,846)	(1,903)	(1,963)	(2,026)	(2,090)	(2,157)	(2,227)	(2,299)
Cashflows from Financing Activities										

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	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Capital	472,871									
Loan Amortization	(117,392)	(128,521)	(140,704)	(154,043)	(168,646)	0	0	0	0	0
Bank Interest Rate	(67,242)	(56,114)	(43,930)	(30,591)	(15,988)	(0)	(0)	(0)	(0)	(0)
Loans	709,307	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	997,543	(184,634)	(184,634)	(184,634)	(184,634)	0	0	0	0	0
Net (Increase/Decrease) in Cash	206,481	15,479	17,942	20,324	22,617	209,441	212,391	215,296	218,149	220,941
Cashflows at the Beginning of Period	0	206,481	221,960	239,901	260,225	282,842	492,283	704,674	919,970	1,138,119
Cashflows at the End of Period	206,481	221,960	239,901	260,225	282,842	492,283	704,674	919,970	1,138,119	1,359,060

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 36 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	206,481	221,960	239,901	260,225	282,842	492,283	704,674	919,970	1,138,119	1,359,060
Inventory	24,500	25,235	25,992	26,772	27,575	28,402	29,254	30,132	31,036	31,967
Accounts Receivable	0	0	0	0	0	0	0	0	0	0
Total current Assets	230,981	247,195	265,893	286,997	310,417	520,685	733,928	950,102	1,169,155	1,391,027
Non Current Assets										
Fixed Assets (net)	942,487	889,331	836,047	782,629	729,075	675,381	621,542	567,554	513,414	459,116
Total Non Current Assets	942,487	889,331	836,047	782,629	729,075	675,381	621,542	567,554	513,414	459,116
Total Assets	1,173,468	1,136,526	1,101,940	1,069,626	1,039,492	1,196,066	1,355,470	1,517,656	1,682,569	1,850,143
Liabilities										
Current Liabilities										
Paybles	41,310	42,874	44,497	46,184	47,935	49,754	51,643	53,605	55,643	57,759
Remaining amount of Loan	128,521	140,704	154,043	168,646	(0)	(0)	(0)	(0)	(0)	0

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Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total current Liabilities	169,831	183,578	198,541	214,830	47,935	49,754	51,643	53,605	55,643	57,759
Non Current Liabilities										
Long Terms Loans	463,394	322,690	168,646	0	0	0	0	0	0	0
Total Long Term Liabilities	463,394	322,690	168,646	0	0	0	0	0	0	0
Total Liabilities	633,225	506,268	367,187	214,830	47,935	49,754	51,643	53,605	55,643	57,759
Owners Equity										
Shareholders Contributions	472,871	472,871	472,871	472,871	472,871	472,871	472,871	472,871	472,871	472,871
Statutory Reseve	6,737	15,739	26,188	38,193	51,869	67,344	83,096	99,118	115,405	131,951
Retained Profits	60,635	141,649	235,694	343,733	466,817	606,097	747,861	892,062	1,038,649	1,187,562
Total Equity	540,243	630,259	734,753	854,796	991,557	1,146,312	1,303,827	1,464,051	1,626,926	1,792,384
Total Liabilities and Equity	1,173,468	1,136,526	1,101,940	1,069,626	1,039,492	1,196,066	1,355,470	1,517,656	1,682,569	1,850,143

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 37 Free Net Cash flows Table

Net cash flows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(1,182,178)	215,817	200,113	202,576	204,958	207,251	209,441	212,391	215,296	218,149	220,941	3,020,897
Discount Factor	1.00	0.80	0.72	0.64	0.58	0.52	0.46	0.42	0.37	0.33	0.30	0.30
Net Present Value for Cash Flows	(1,182,178)	173,264	143,949	130,567	118,365	107,242	97,105	88,232	80,138	72,755	66,024	902,733

Table below (42) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 38 Payback Period

Net Free Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project Value Returned	(1,182,178)	215,817	200,113	202,576	204,958	207,251	209,441	212,391	215,296	218,149	220,941
Payback Period (Year)	1,182,178	966,361	766,248	563,672	358,714	151,463	(57,978)	(270,369)	(485,665)	(703,814)	(924,755)
Net Free Cashflows	5	1	1	1	1	1	0	0	0	0	0

Table 39 Financial Analysis Results

Indicator	Unit measure	Value
Net Present Value for Cashflows	Ratio	%11.61
Payback Period	JOD	798,194
Internal Rate of Return	Year	5
Net Present Value for Cashflows	Ratio	%21.58

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

The following table shows that the project remains economically profitable even if revenue declined by 10%, or rise operating expenses by 10%.

Table 40 Sensitivity Analysis

Sensitivity analysis	IRR	Payback period	WACC
Original Scenario	%21.58	5	%11.61
Revenues declined by 10%	%16.51	7	%11.61
Operating Expenses Increased by 10%	%15.33	7	%11.61

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 798,194 Jordan Dinars considering that the project provides 39 Job Opportunities for the governorate residents.